Foundations of Photonic Quantum Computing: A Pedagogical Introduction

Satchith P.S (Mentor)*[1], Abhishek P.G†[2], and Sidharth.T.Nandiath‡[3]

[1]*Department of Physics, Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu - 641112, India*
[2]*Department of Physics, St. Aloysius College, Elthuruth, Thrissur, Kerala - 680611, India*
[3]*Department of Physics, Indian Institute of Technology, Hyderabad, Kandi, Sangareddy, Telangana - 502285, India*

Abstract

Photonic quantum computing harnesses the quantum properties of light to process and transmit information in fundamentally new ways. This review introduces the foundational concepts of the field with an emphasis on pedagogy and accessibility. We outline how qubits can be encoded in photons through various schemes, followed by a discussion of basic optical components that serve as quantum gates. We also examine entanglement generation and quantum interference, highlighting processes such as spontaneous parametric down-conversion and the Hong–Ou–Mandel effect. By focusing on conceptual clarity rather than mathematical formalism, this review article provides an entry point for students and early-stage researchers into the foundations of photonic quantum computing.

Keywords: Photonic quantum computing, Quantum information science, Qubits, Encoding, Linear optical quantum computing (LOQC), Quantum interference, Hong–Ou–Mandel effect, Spontaneous parametric down-conversion (SPDC), Pedagogical review

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*ps.satchith@gmail.com
†abhishekpgireesh@gmail.com
‡sidharthtnandiath@gmail.com

1 Introduction

From the earliest days of civilization, humans have relied on calculations and problem solving to meet their needs. Whether it was keeping track of resources or predicting the movement of celestial bodies, much of human progress has been built upon our ability to compute. This long history of calculation eventually led to the invention of computers.

Computation as a concept can be traced back to early civilizations such as the Babylonians and gradually developed through the work of great thinkers, most notably Alan Turing [1]. In 1936, Turing demonstrated that any algorithm could, in principle, be executed on a single universal machine, an idea that laid the foundation for modern computer science.

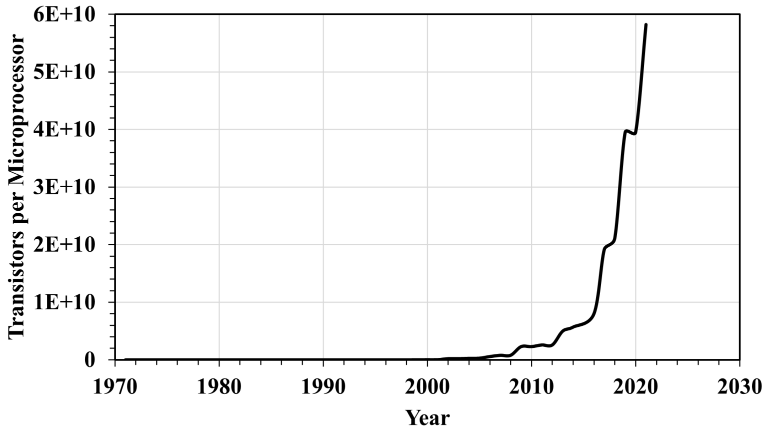


Figure 1: The Moore's law shows the exponential plot of the number of transistors per microprocessor over the years

For decades, advancements in computing closely followed Moore's law, which predicted that computing power would double as transistors became increasingly smaller (Illustrated in Fig. 1). This trend continued for many years. However, as transistors reached the scale of individual atoms, quantum effects began to disrupt their functioning, limiting the traditional hardware design. Faced with this barrier, researchers turned to new approaches, ranging from probabilistic models to the development of quantum systems that process information in fundamentally different ways.

Quantum computation, as a field, explores how quantum mechanical systems can be used to perform information processing tasks that classical computers struggle with [2]. Unlike conventional machines that rely on binary logic, quantum computers draw power from phenomena such as *superposition*, *entanglement*, and *interference*. These are not just theoretical notions, but real physical resources that provide quantum computers with their extraordinary capabilities.

1.1 Classical vs Quantum Computers

To understand how quantum computers differ from the traditional classical computers, let us first compare them with classical computers. The key differences are highlighted in Table 1[**article**]:

Sl. No.	Classical Computer	Quantum Computer
1	Uses **bits** as the basic unit of information, which can be 0 or 1.	Uses **qubits** as the basic unit of information, which can exist in a superposition of $\|0\rangle$ and $\|1\rangle$.
2	Computation is **deterministic**: the same input always produces the same output.	Computation is **probabilistic**: measurement outcomes follow quantum probabilities.
3	Information is processed using **logic gates** like AND, OR, NOT.	Information is processed using **quantum gates** such as Hadamard, Pauli-X, and CNOT.
4	Scaling is limited by Moore's law and transistor miniaturization.	Scaling exploits quantum properties like superposition, entanglement, and interference.
5	Well-suited for everyday tasks and general-purpose computing.	Specialized for certain problems, e.g., factoring, search, and simulating quantum systems.

Table 1: Comparing a Classical Computer and a Quantum Computer

A quantum computer uses qubits as its fundamental unit of information. Unlike classical bits, quantum bits or qubits can exist in superpositions of 0 and 1, that is, they can represent multiple states simultaneously. This allows quantum computers to explore many possible solutions in parallel (a concept called quantum parallelism), rather than one at a time, thus providing solutions much faster. Moreover, quantum mechanics introduces phenomena such as entanglement, which links qubits in such a way that the state of one qubit is instantly related to another, no matter the distance. Quantum interference is then used to amplify the desired answers while canceling out unwanted ones, giving quantum algorithms their unique advantage.

So one may think *"This is great! Then why do we require classical computers when quantum computers outperform the classical ones by orders of magnitude?"* But this is not the case.

While quantum computers can indeed provide exponential or quadratic speedups for certain specialized problems, they are not universally superior. For most everyday tasks - such as browsing the Internet, running spreadsheets, or editing documents- classical computers already perform extremely well. These tasks are solved efficiently by algorithms that fall into class **P** (Polynomial), and a quantum computer would offer no real advantage. Quantum computers work probabilistically: they give results according to probability distributions, which means several runs are often required to achieve reliable outcomes. This makes them less suitable for routine applications where accuracy and consistency are critical.

The following paragraphs would give a more intuitive introduction to a person from a computer science background. It is explained in a way that is easily grasped by all readers. One may skip this and proceed to Sec. 1.2 in case of difficulty understanding it.

The class of problems that can be solved in polynomial time by a classical computer is called

P [3]. These are the problems for which classical computers are considered more efficient than a quantum computer.

Another important class is **NP** (Nondeterministic Polynomial time). These are problems where a proposed solution can be verified in polynomial time by a deterministic classical computer [4]. The famous **P** vs **NP** problem asks whether every problem whose solution can be checked quickly can also be solved quickly in the first place, a question still unresolved. Problems such as the traveling salesman or satisfiability (SAT) belong to this realm, and many are believed to be much harder than anything in **P**[4].

Quantum computation introduces a new paradigm. A quantum computer uses qubits instead of bits, and these qubits can exist in superpositions of 0 and 1. Furthermore, qubits can be entangled, creating correlations that have no classical counterpart. Computation is performed by applying unitary transformations (quantum gates) and is finalized by measurement. The class of problems solvable efficiently by such machines is known as **BQP** (Bounded-error Quantum Polynomial time) [2]. 'Bounded error' refers to the fact that quantum algorithms do not always give the right answer with certainty, but can amplify the probability of correctness to remarkably high levels. In terms of relationships,

$$\mathbf{P} \subseteq \mathbf{BQP} \subseteq \mathbf{PSPACE}$$

where, **PSPACE** is the class of decision problems that can be solved using only a polynomial amount of memory (space), regardless of how much time it takes.

It is widely believed that quantum computers cannot solve all **NP-complete** problems efficiently, but they can efficiently solve certain problems that classical machines cannot, like the famous Shor's Algorithm(finding prime factors of large integers) [5] and the Grover's Algorithm(which speeds up unstructured search problems quadratically) [6].

Therefore, it is not accurate to say that one approach is entirely better than the other. Each has particular strengths that make it useful for different types of problems. In fact, current research suggests that hybrid systems, those combining classical and quantum methods, are likely to play a central role in the near future [7].

1.2 Qubits and Bloch Sphere Representation

We have already seen that a **qubit** is the fundamental unit of a quantum computer. Unlike a classical bit, which can only be 0 or 1, a qubit can exist in a linear combination of both states at the same time. Let us formally introduce this concept. Mathematically, a qubit is expressed as [8]:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \tag{1}$$

where α and β are complex probability amplitudes that satisfy $|\alpha|^2 + |\beta|^2 = 1$. For example, the polarization state of a photon in a superposition of two states $|H\rangle$ and $|V\rangle$ with equal probabilites can be expressed as-

$$|\psi\rangle = \frac{1}{\sqrt{2}} |H\rangle + \frac{1}{\sqrt{2}} |V\rangle$$

This unique property allows quantum computers to perform computations in parallel and is the foundation of quantum algorithms.

To better visualize this, the state of a single qubit can be represented on a geometric model called the **Bloch sphere** (refer to Fig. 2 for illustration).

Any qubit state can be written as[2]

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \tag{2}$$

where θ and ϕ are real numbers that correspond to angles on the surface of a unit sphere. representing the latitude and longitude of the sphere respectively. This way, the Bloch sphere provides an intuitive way to understand qubits:

The north pole corresponds to $|0\rangle$, the south pole corresponds to $|1\rangle$, and any other point on the sphere corresponds to a superposition of $|0\rangle$ and $|1\rangle$.

This geometric picture is useful because it shows how quantum gates correspond to the rotations of the qubit state on the Bloch sphere.

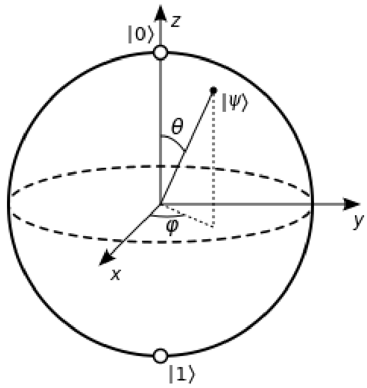


Figure 2: Bloch sphere representation of a single qubit. The north pole corresponds to $|0\rangle$ and the south pole to $|1\rangle$.

1.3 Quantum Gates

The basic function of a quantum gate is to change the state of a qubit from one to another. Mathematically, it denotes the linear transformation of a vector or a matrix multiplication. Physically, they can be implemented using linear optical elements that perform specific operations on the qubits. Some important gates and their implementations will be discussed in the following sections. References have been taken mainly from [2, 9, 10, 11, 12, 13].

1.3.1 Pauli-X Gate

This gate performs a bit-flip operation on a qubit, i.e. changes input state $|0\rangle$ to $|1\rangle$ and vice-versa. It is analogous to classical **NOT** gate, hence known as **quantum NOT** gate or *bit-flip* gate. Due to the involvement of only a single qubit, this is a single-qubit gate. On a Bloch sphere, it can be visualized as performing a rotation around the x-axis by $\boldsymbol{\pi}$ radians. It's matrix representation is as follows:

$$\boldsymbol{\sigma_x} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{bmatrix}$$

The Pauli-X gate can be physically realized by orienting a half-wave plate at 45°, thus flipping the states between $|H\rangle$ or $|V\rangle$ respectively.

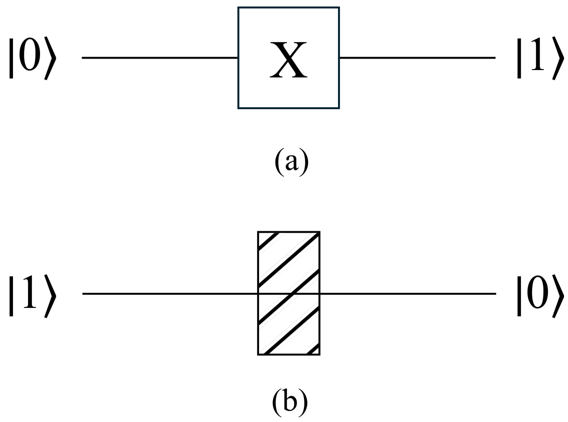


Figure 3: (a) Symbol for X-Gate (b) HWP at 45°

1.3.2 Pauli-Z Gate

The Pauli-Z Gate performs a phase-flip rotation on a qubit, hence also known as a *phase-flip* gate. It can be visualized as performing a rotation around the z-axis of Bloch sphere by $\boldsymbol{\pi}$radian. It essentially transforms the state $|+\rangle$ to $|-\rangle$ and vice-versa. Its matrix representation is given as-

$$\boldsymbol{\sigma_z} = \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix}$$

For polarization encoding, $\sigma_z |H\rangle = |H\rangle$ and $\sigma_z |V\rangle = -|V\rangle$.

This gate can be physically realized using a phase shifter that adds a phase of π only to state $|V\rangle$. For example, if a half-wave plate is oriented with slow axis aligned vertically, a 180° phase shift to $|V\rangle$ basis. HWP at 0° also implements a Pauli-Z gate.

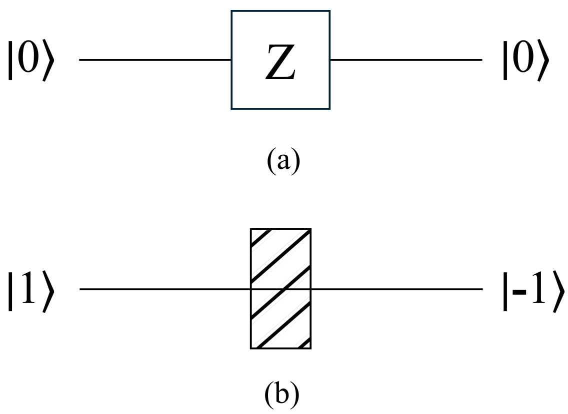


Figure 4: (a) Symbol for Z-Gate (b) HWP oriented at 0°

1.3.3 Pauli-Y Gate

This gate performs both a *bit-flip* and a *phase-flip* rotation on a qubit. So, it is basically a combination of Pauli-X and Z gates. It can be visualized as performing a rotation around the y-axis of Bloch sphere by π radian. The matrix representation is given as-

$$\boldsymbol{\sigma_y} = \begin{bmatrix} \mathbf{0} & -\boldsymbol{i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix} = \boldsymbol{i\sigma_x\sigma_z}$$

In the case of polarization encoding, $\sigma_y |H\rangle = i |V\rangle$ and $\sigma_y |V\rangle = -i |H\rangle$. It can be physically realized by combining the above Pauli-X and Pauli-Z gates, i.e. a HWP at 45° followed by a HWP with slow axis aligned vertically.

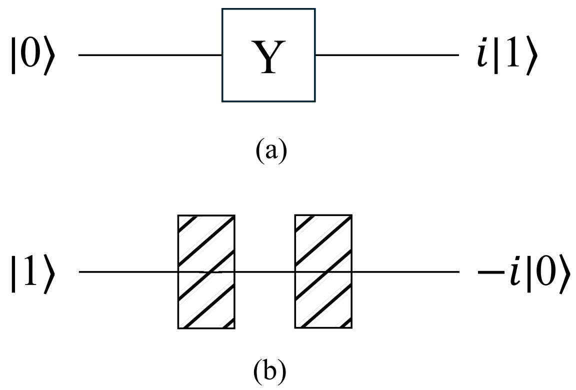


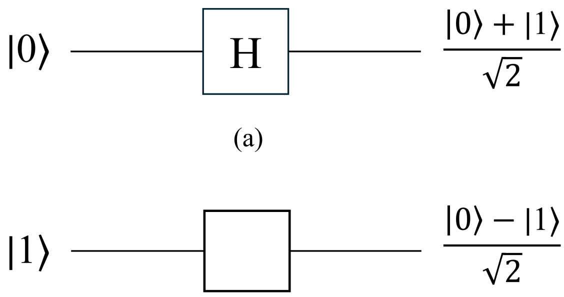
Figure 5: (a) Symbol for Y-Gate (b) HWP oriented at 45° followed by one at 0°

1.3.4 Hadamard Gate

This is one of the most widely used and important gates in quantum computing. This single-qubit operation converts the z-basis $|0\rangle, |1\rangle$ to x-basis $\{|+\rangle, |-\rangle\}$ and vice-versa. It essentially creates a superposition state of basis states $|0\rangle$ and $|1\rangle$. The Hadamard gate is represented by the matrix-

$$H = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}} \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & -\mathbf{1} \end{bmatrix}$$

For the polarization encoding of photons, Hadamard gate acting on state $|H\rangle$ results in $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ and one acting on state $|V\rangle$ results in $\frac{1}{\sqrt{2}}(|H\rangle - |V\rangle)$. It can be physically implemented by orienting a H.W.P at 22.5°. In the case of path-encoding, a 50:50 beam-splitter acts as a Hadamard gate. Note that the input must be a single-photon source.



(b)

Figure 6: (a) Symbol for Hadamard Gate (b) HWP at 22.5°(for polarization encoding) or a 50:50 BS(for path encoding).

1.3.5 Rotation Gates

The rotation operator gates $R_x(2\theta)$, $R_y(2\theta)$ and $R_z(2\theta)$ represent the transformations of the x, y and z axes of the Bloch sphere. For n $\in \{x, y, z\}$, they can be mathematically represented as-

$$\boldsymbol{R_n(2\theta) = e^{-in\theta}}$$

In matrix notation,

$$\boldsymbol{R_x(2\theta) = \begin{bmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{bmatrix}}$$

$$\boldsymbol{R_y(2\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}$$

$$\boldsymbol{R_z(2\theta) = \begin{bmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{bmatrix}}$$

Placing a half-wave plate at θ in between two quarter-wave plates each at 0° physically realizes an x-rotation gate:

$$R_x(2\theta) = QWP(0°) \cdot HWP(\theta) \cdot QWP(0°)$$

Here, HWP controls the rotation angle. Implementing the z-rotation gate requires a HWP to be sandwiched between two QWPs at 45° each:

$$R_z(2\theta) = QWP(-45°) \cdot HWP\left(\frac{\theta}{2}\right) \cdot QWP(45°)$$

Realizing the y-rotation gate using optical elements is relatively easier as we just require a single half-wave plate:

$$R_y(2\theta) = HWP(\theta)$$

The rotation gate, an important gate for initializing qubit states is easily realizable with just waveplates. We shall see its implementation in the coming sections.

1.3.6 CNOT Gate

The **CNOT** gate is a fundamental 2-qubit gate which is the quantum analogue of the classical **XOR** gate. If the first qubit (also known as the '*control*' qubit) in state $|0\rangle$ or $|H\rangle$ is acted on by this gate, then the output remains unchanged. But if the first qubit is in the state $|1\rangle$ or $|V\rangle$, then the state of second qubit (also known as the '*target*' qubit) will be flipped. The **CNOT** gate is also known as a **controlled NOT** or **controlled X** or **CX** gate. The matrix representation of the CNOT gate is as follows:

$$\boldsymbol{CNOT} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix}$$

A convenient approach of physically realizing the CNOT gate is by combining polarization-encoding $\{|H\rangle, |V\rangle\}$ and path-encoding (two arms of an interferometer) of a single photon. Both the qubit states can be manipulated independently by changing either the polarization(rotation) or the path state (phase shift in one arm of interferometer). Usually, the path state is considered as control qubit which can flip the polarization state of second(target) qubit.

First, passing the beam through 50:50 beam-splitter or a polarizing beam-splitter which creates a superposition state of photon being in transmitted path($\equiv |0\rangle$) and reflected path($\equiv |1\rangle$). Since we require the photon in reflected path to flip its polarization state, we insert a HWP aligned 45° to path $|1\rangle$, thus creating a CNOT gate.

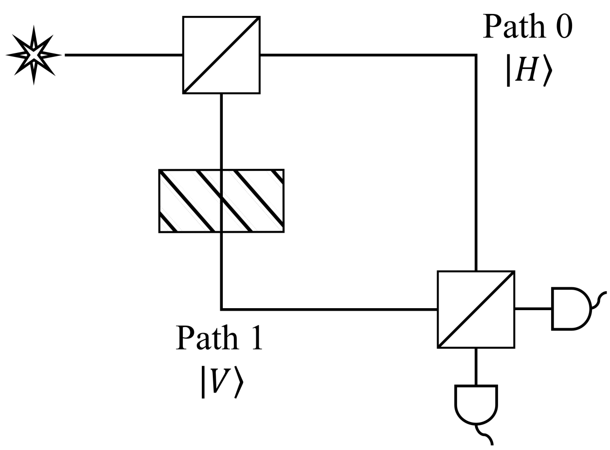


Figure 7: The CNOT gate. A HWP (at 45°) is inserted into the arm of interferometer which corresponds to the $|V\rangle$ state. The final state is measured by the detectors placed at the output ports of the second BS.

Note that the above construction uses the path of a single photon as a control and its polarization as the target, that is, two degrees of freedom of the same particle. This is not a deterministic two-photon entangling CNOT; entangling two separate photonic qubits deterministically typically requires ancilla photons, projective measurements, or strong nonlinearities (see [11, 14]).

1.4 Physical Realization of Qubits

Even though qubits are introduced as mathematically abstract entities, they must ultimately be implemented in physical systems to build quantum computers [**article**, 2] . Due to the fact that they have two basis states, several different physical systems have been explored for realizing qubits such as:

- **Energy Level of Atoms:** The 'ground' $|g\rangle$ and 'excited' $|e\rangle$ energy states of an atom.
- **Spin-$\frac{1}{2}$ Particles:** The spin-up $|\uparrow\rangle$ and spin - down $|\downarrow\rangle$ of a spin-$\frac{1}{2}$ particle like electrons, protons, neutrons, etc.
- **Harmonic Oscillators:** Qubits encoded in the lowest two levels of an oscillator, such as a cavity mode.
- **Superconducting Qubits:** The two lowest energy states of stable Josephson junctions in superconducting circuits.
- **Trapped Ions:** Qubits encoded in the internal states of ions confined using electromagnetic fields.
- **Spin Qubits:** Based on electron or nuclear spin states in quantum dots or defects in solids.
- **Photonic Qubits:** Qubits represented by photon polarization, path, orbital angular momentum, or time-bin degrees of freedom.

Each of these systems provides a physical understanding of the abstract qubit, and current research continues to investigate which technologies are best suited for large-scale quantum computing. This work mainly focuses on photonic qubits - understanding them and ultimately realizing them using linear optical elements.

1.5 Photons and Decoherence

So what is a *photon*? This in itself is a whole different story, but let us stick to our current understanding of this entity. According to quantum electrodynamics (QED), the **photon** is the smallest and *the fundamental unit of excitation* associated with a quantized mode of the radiation field [15]. In layman terms, light can be thought of as made up of a large number of these photons.

Photons are valuable for quantum information only if they maintain a stable **coherent** state. A photon is said to be coherent if its quantum state maintains a well-defined phase relationship over time and space, allowing for interference phenomena that are key to quantum technologies [16]. Temporal coherence is a measure of the correlation of a light wave (or any wave) with itself at different points in time (e.g., single-photon interference in a Michelson interferometer), while spatial coherence determines the visibility of interference patterns across different paths, i.e. tells us whether two points on a wavefront (in space) can interfere with each other. The correlation basically tells us how much two quantities vary together [17].

When qubits maintain coherence, they can store and process information in ways not possible with classical bits. But when the system interacts with the environment, coherence gets disturbed, thus resulting in information being lost. This process is called **decoherence** [18]. A photon has various degrees of freedom and interacts very weakly with its environment, thereby maintaining the coherence that is required throughout the quantum computation. Also note that photons travel at very high speeds and can operate in room temperatures, making them an ideal carrier of information [19].

Thus, coherence is key to quantum computation, whereas decoherence is one of the biggest challenges in building real quantum computers. The points outlined above provide the motivation for investigating photonic quantum systems in this work.

2 Encoding Qubits in Photons

Quantum information processing requires physical systems that can store and manipulate quantum bits (qubits). While several physical implementations exist, such as trapped ions, superconducting circuits, and spins in semiconductors- photons stand out as particularly promising qubit carriers. The reasons are straightforward: photons are fast, interact weakly with the environment (minimizing decoherence), and can be transmitted over long distances with relatively low loss. Furthermore, photons are already at the heart of modern communication technologies, making them natural candidates for quantum networks and quantum communication protocols.

3 Photonic Qubit Encodings

What gives photonic qubits an edge over the others is that they possess multiple degrees of freedom (DoFs)—such as polarization, path, time of arrival, frequency, and orbital angular momentum—that can serve as qubit encodings [10, 20]. Each DoF can be used independently or in combination, offering a rich toolbox for implementing various quantum tasks. In this section, we explore these encoding methods in detail, comparing their strengths, limitations, and areas of application.

3.1 Polarization Encoding

Polarization is the most familiar method of encoding qubits in photons [21]. The two logical basis states are defined as:

$$|0\rangle \equiv |H\rangle\,, \quad |1\rangle \equiv |V\rangle\,,$$

where $|H\rangle$ and $|V\rangle$ denote horizontal and vertical polarizations, respectively. Alternatively, circular polarizations $(|L\rangle, |R\rangle)$ may serve as basis states.

Polarization qubits are easy to prepare with polarizers and half-wave plates, manipulated with additional wave plates or electro-optic devices, and measured using polarizing beam splitters (PBS) combined with single-photon detectors. Superposition states such as

$$|\psi\rangle = \alpha\,|H\rangle + \beta\,|V\rangle$$

are straightforwardly realized by rotating the polarization axis of the photon.

3.2 Path (Dual-Rail) Encoding

In path or dual-rail encoding, a single photon is placed in one of two distinct spatial modes [14]. Logical states are defined as:

$$|0\rangle \equiv \text{photon in mode A}, \quad |1\rangle \equiv \text{photon in mode B}.$$

A balanced beam splitter can create coherent superpositions of these two paths:

$$|0\rangle \to \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Manipulation of path qubits is achieved using mirrors, beam splitters, and phase shifters, while detection is performed at the respective output ports. Path encoding is particularly well suited to interferometer-based setups and integrated photonic circuits.

3.3 Time-Bin Encoding

Time-bin encoding defines qubit states by the photon's arrival time [22]. An unbalanced interferometer can split a photon wave packet into two temporal modes:

$$|0\rangle \equiv \text{“early” time bin}, \quad |1\rangle \equiv \text{“late” time bin}.$$

Superpositions are created by carefully balancing the optical paths so that a photon exists simultaneously in early and late modes:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle).$$

Time-bin qubits are typically generated and measured using interferometers combined with phase modulators and fast single-photon detectors.

3.4 Frequency (Energy-Bin) Encoding

In frequency encoding, different frequency components of a photon represent logical states [23]:

$$|0\rangle \equiv |\omega_1\rangle, \quad |1\rangle \equiv |\omega_2\rangle,$$

where ω_1 and ω_2 denote distinct frequencies. A photon can be placed in a superposition of these two frequency modes:

$$|\psi\rangle = \alpha|\omega_1\rangle + \beta|\omega_2\rangle.$$

Preparation and manipulation are accomplished with electro-optic modulators, Bragg gratings, or micro-ring resonators, while measurement is performed by filtering the output into the corresponding frequency bins. Frequency encoding is especially relevant in multiplexed quantum communication.

3.5 Orbital Angular Momentum (OAM) Encoding

Photons can also carry orbital angular momentum (OAM), associated with their helical phase structure. Distinct OAM modes, labeled by integers ℓ, define the qubit basis [24, 25]:

$$|0\rangle \equiv |\ell_1\rangle, \quad |1\rangle \equiv |\ell_2\rangle.$$

OAM states can be generated using spatial light modulators or q-plates, and superpositions of modes are realized by shaping the optical wavefront appropriately. Measurement is performed with OAM mode sorters, which spatially separate different OAM states before detection.

Unlike polarization, OAM states extend naturally to higher-dimensional systems (qudits), allowing encoding in Hilbert spaces of dimensions larger than two. This makes OAM particularly powerful for exploring high-capacity quantum information protocols.

3.6 Hybrid and Multi-DoF Encodings

Beyond single-DoF encodings, photons can be encoded simultaneously in multiple degrees of freedom. For example, polarization–path, polarization–time bin, or path–frequency encodings allow richer information capacity and flexibility [26]. Such hybrid encodings enable compact experimental designs, the preparation of high-dimensional entangled states, and the implementation of more complex protocols on fewer photons. Multi-DoF approaches are especially useful for generating photonic cluster states and for studying error correction, as they provide redundancy and enhanced resilience against noise.

3.7 Comparative Summary

The different encoding methods are summarized in Table 2. Each method uses distinct degrees of freedom of photons and provides a natural way to define logical qubit states and their superpositions.

Thus, encoding qubits in photons provides a diverse and powerful tool for quantum technologies. Polarization is conceptually simple and widely used, path encoding integrates naturally with photonic circuits, time-bin encoding excels in fiber-based communication, frequency encoding supports multiplexing, and OAM enables exploration of higher-dimensional Hilbert spaces.

Future progress will likely rely on combining these approaches in hybrid or multi-DoF architectures, which can exploit the advantages of each encoding scheme simultaneously. Such developments are crucial for building scalable quantum networks, implementing fault-tolerant photonic quantum computing, and realizing practical applications of quantum information science.

Sl.No.	Degree of Freedom	$\|\mathbf{0}\rangle$	$\|\mathbf{1}\rangle$	Applications
1	Polarization	$\|H\rangle$	$\|V\rangle$	Free-space QKD, teleportation demonstrations, Bell tests, lab-scale gate demonstrations
2	Path (Dual-Rail)	Photon in mode A	Photon in mode B	Linear optical quantum computing (LOQC), boson sampling, interferometer-based algorithms
3	Time-Bin	Early arrival	Late arrival	Fiber-based QKD, long-distance entanglement distribution, telecom quantum links
4	Frequency (Energy-Bin)	$\|\omega_1\rangle$	$\|\omega_2\rangle$	Multiplexed quantum channels, quantum networking, interface to frequency-domain memories
5	Orbital Angular Momentum (OAM)	$\|\ell_1\rangle$	$\|\ell_2\rangle$	High-dimensional QKD, free-space channels with increased capacity, quantum imaging
6	Hybrid / Multi-DoF	Combination	Combination	Cluster states, compact multi-qubit experiments, entanglement swapping and complex protocols

Table 2: Degree of freedom, logical states, and applications.

4 Optical Elements as Quantum Gates

Quantum computers can be built on several physical platforms, including superconducting qubits, trapped ions, photonic qubits, neutral atoms, and topological qubits [27, 28]. Among these, photonic quantum computing offers a particularly promising approach [19].

Classical optics explains many properties of light but does not capture its full behavior. Light also shows unique quantum features, such as superposition, entanglement, and interference, which are central to quantum optics [16]. These properties are exactly the resources used in quantum computing. Photons are especially well-suited for carrying quantum information because they move at the speed of light and are less affected by environmental noise compared to other systems. This makes them ideal not only for computation, but also for quantum communication and networking. However, photons interact only weakly with each other, so advanced tools are needed to manipulate and control them [19].

In photonic quantum computing, quantum information can be encoded in various photon properties such as polarization, path, arrival time, or orbital angular momentum. To control these qubits, optical elements such as beam splitters, mirrors, and phase shifters are used. These are not just supporting components but the essential building blocks that allow photonic qubits to be generated, manipulated, and measured [14].

The process begins with photon generation, which is achieved using single-photon sources.

Unlike classical light, which emits many incoherent photons, quantum computers require photons that are indistinguishable and emitted on demand. This ensures that interference effects, which are crucial for quantum logic operations, can occur reliably. Without indistinguishable photons, large-scale photonic quantum computing would not be possible [29].

A widely used method to generate single photons is a process known as spontaneous parametric down-conversion (SPDC), where a high-energy photon interacts with a nonlinear crystal and splits into two lower-energy photons (signal and idler). These are produced in an entangled state, which is a vital resource for quantum algorithms [30]. A more modern approach uses quantum dots, which are nanoscale semiconductors that emit a single photon when excited by a laser pulse. Quantum dots allow for greater control over the timing and properties of photon emission [31].

Ultimately, the reliability of photonic quantum computing depends on these high-quality photon sources. They provide the starting point of any photonic quantum circuit, since without them the accuracy of the entire computation would be compromised. Let us understand the working of various optical elements and how computation can be performed using them.

4.1 Mirrors

Mirrors are one of the simplest yet most essential optical elements in photonic quantum computing. Unlike wave plates or beam splitters, mirrors do not typically perform a direct logical gate operation on a photonic qubit. Instead, they are used to redirect the path of photons, thereby enabling the construction of interferometers and multi-path quantum circuits. Hence, they simply "guide" qubits between different stages of computation [32, 33].

From a quantum-mechanical perspective, the action of a mirror can be represented as a unitary transformation in the spatial mode of a photon. For a perfect mirror, the primary effect is to reflect the incoming mode onto a new direction while adding a global phase factor. A single reflection can be written as:

$$|\psi_{in}\rangle \xrightarrow{M} e^{i\phi} |\psi_{out}\rangle$$

where, $|\psi_{in}\rangle$ is the incident photonic state, $|\psi_{out}\rangle$ is the reflected state, and $e^{i\phi}$ is the phase introduced by reflection (commonly $\phi = \pi$, though it depends on the mirror's coating and polarization).

In the case of path-encoded qubits, where the logical basis states are defined as $|0\rangle \equiv$ photon in the upper arm and $|1\rangle \equiv$ photon in the lower arm, a mirror preserves the computational basis while possibly introducing a relative phase. For example, a two-mode system with basis $\{|0\rangle, |1\rangle\}$ can be described by the unitary matrix:

$$M = \begin{bmatrix} e^{i\phi_0} & 0 \\ 0 & e^{i\phi_1} \end{bmatrix}$$

where ϕ_0 and ϕ_1 are the reflection-induced phases on the two possible paths. This representation shows that mirrors do not mix the basis states but can adjust their relative phases, which becomes highly significant in interferometric setups.

4.2 Polarizers

A linear polarizer selectively passes the unpolarized electric field vibrations along a preferential direction, while removing vibrations in the perpendicular direction to be transmitted [34, 35]. In the diagram below, unpolarized light travelling in +z direction passes through linear polarizer, whose transmission axis (T.A) is vertical. Only the component of electric field along the T.A direction is transmitted, i.e. linearly polarized in vertical direction whereas the horizontal component is removed by absorption.

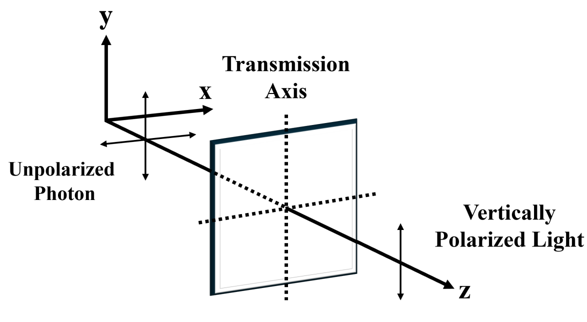


Figure 8: A Linear Polarizer with vertical transmission axis

4.3 Phase Retarders

4.3.1 Half-Wave Plate(HWP)

A half-wave plate is one of the most widely used optical elements for manipulating photon polarization, which is a common encoding scheme for qubits in photonic quantum computing. In polarization encoding, the logical basis states are defined as $|0\rangle \equiv |H\rangle$ (horizontal polarization) and $|1\rangle \equiv |V\rangle$ (vertical polarization). A half-wave plate introduces a relative phase shift of π (half a wavelength) between the fast and slow axes of the crystal.

The action of a **HWP** can be represented by a unitary operator that depends on the orientation angle θ of the plate's fast axis with respect to the horizontal polarization. The transformation matrix in the $\{|H\rangle, |V\rangle\}$ basis is [35]:

$$U_{\mathrm{HWP}}(\theta) = \begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}.$$

At specific angles, the HWP performs important quantum gate operations. For example, at $\theta = 45^\circ$, it swaps horizontal and vertical polarizations, acting as a Pauli-X (NOT) gate. At $\theta = 22.5^\circ$, it maps $|H\rangle$ into an equal superposition of $|H\rangle$ and $|V\rangle$, effectively realizing the Hadamard gate (up to a global phase). In this way, the HWP provides a flexible method for implementing single-qubit gates by simple mechanical rotation. Note that we follow this optical convention, as all single-qubit gate statements are up to an unimportant global phase.

4.3.2 Quarter-Wave Plate(QWP)

A quarter-wave plate introduces a relative phase shift of $\pi/2$ (a quarter of a wavelength) between its fast and slow axes [35]. This element is essential for converting linear polarization into circular polarization and vice versa. This corresponds to introducing relative phase differences between the logical states.

The transformation matrix of a QWP at orientation angle θ in the $\{|H\rangle, |V\rangle\}$ basis is given by:

$$U_{\text{QWP}}(\theta) = \begin{bmatrix} \cos^2\theta + i\sin^2\theta & (1-i)\cos\theta\sin\theta \\ (1-i)\cos\theta\sin\theta & \sin^2\theta + i\cos^2\theta \end{bmatrix}.$$

For example, at $\theta = 45°$, the QWP transforms $|H\rangle$ into the right-circularly polarized state $|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle)$. Similarly, it transforms $|V\rangle$ into the left-circularly polarized state $|L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle)$. In terms of quantum gates, the QWP can act as a phase gate or a rotation around the Bloch sphere's Z-axis, depending on its orientation.

Quarter-wave plates are especially important in interferometric experiments, where they allow precise control over the relative phases of quantum states and enable the creation of arbitrary single-qubit rotations when combined with half-wave plates.

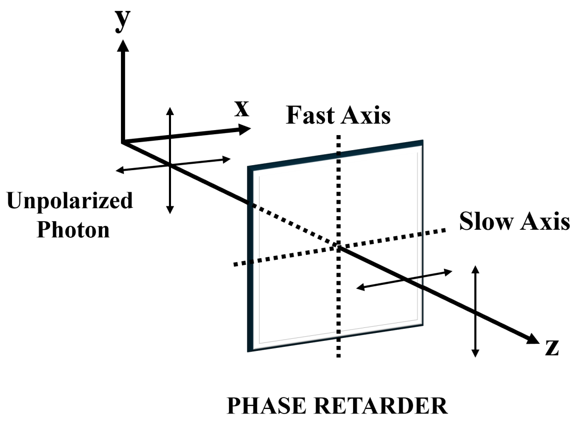


Figure 9: The phase retarder induces a phase difference on the two polarized components, when the unpolarized light passes through it.

4.4 Beam-Splitters(BS)

Generally, a beam-splitter (BS) is made from thin metal or dielectric layers on glass and splits an incoming beam into two: a reflected beam and a transmitted beam. The most useful application of a beam splitter is an interferometer which is designed to split a single light beam and then recombine the two resulting beams. Reference has been taken from [36].

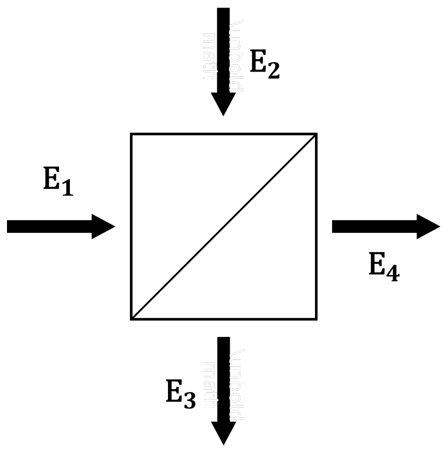


Figure 10: A Classic beam-splitter with input ports 1, 2 and output ports 3,4.

When a photon hits the beam-splitter, the photon does not split into two halves. Instead, it has an equal probability to be either reflected or transmitted. This creates a superposition of states, which is a key concept in quantum mechanics. This is due to the fact that photons are the most fundamental entity and cannot be split further.

A 50:50 beam-splitter is a common type where there is a 50% chance the photon will go one way and a 50% chance it will go the other. Mathematically, we can represent the photon's path with a simple two-element list, like a column vector, where the numbers represent the complex probability amplitude of the photon being in the first and the second paths, respectively. When a photon enters a 50:50 beam-splitter, the convenient unitary convention of representing the situation is a mathematical operation represented as-

$$U_{BS} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$$

Consider a laser light beam incident on one of the input ports of a 50:50 beam-splitter as shown in Figure 10. This electric field incident on the input port, $\boldsymbol{E_1}$ is reflected and transmitted as fields $\boldsymbol{E_3}$ and $\boldsymbol{E_4}$ respectively. Likewise, another field incident on the other input port, $\boldsymbol{E_2}$ is transmitted and reflected as $\boldsymbol{E_3}$ and $\boldsymbol{E_4}$ respectively. The output beams can be expressed as follows:

$$\boldsymbol{E_3 = rE_1 + tE_2}$$
$$\boldsymbol{E_4 = r'E_2 + t'E_1}$$

Here, $\boldsymbol{r}$ and $\boldsymbol{r'}$ are the reflection coefficients whereas $\boldsymbol{t}$ and $\boldsymbol{t'}$ are the transmission coefficients. The matrix representation for above set of equations is given as:

$$\begin{bmatrix} \boldsymbol{E_3} \\ \boldsymbol{E_4} \end{bmatrix} = \begin{bmatrix} \boldsymbol{r} & \boldsymbol{t} \\ \boldsymbol{t'} & \boldsymbol{r'} \end{bmatrix} \begin{bmatrix} \boldsymbol{E_1} \\ \boldsymbol{E_2} \end{bmatrix}$$

Note that for a symmetric beam-splitter, $\boldsymbol{r = r'}$ and $\boldsymbol{t = t'}$. On applying the energy conservation laws,

$$\boldsymbol{|E_1|^2 + |E_2|^2 = |E_3|^2 + |E_4|^2}$$

Also, the following conditions must satisfy:

$$|\boldsymbol{r}|^{\mathbf{2}} + |\boldsymbol{t}|^{\mathbf{2}} = \mathbf{1} \text{ and } \boldsymbol{rt}^* + \boldsymbol{tr}^* = \mathbf{0}$$

The first condition is quite intuitive. The total energy must be conserved, i.e., the square of amplitudes of reflection and transmission coefficients must equal unity. The '$*$' in the second condition denotes 'conjugate transpose'.

4.5 Polarizing Beam-Splitters(PBS)

A polarizing beam splitter (PBS) is a specialized optical element that separates photons according to both their polarization and their spatial path. Unlike a standard beam splitter, which probabilistically divides an incoming beam, a PBS deterministically routes horizontally and vertically polarized photons into different output modes [16]. Typically, horizontally polarized photons $|H\rangle$ are transmitted, while vertically polarized photons $|V\rangle$ are reflected.

This dual functionality means that the PBS acts as both a polarization and a path-routing device. Formally, for input polarization states, the PBS implements the transformations:

$$|H\rangle \longrightarrow |H\rangle_T\,, \qquad |H\rangle \longrightarrow |H\rangle_R\,,$$

where the subscripts T and R denote the transmitted and reflected spatial modes, respectively. Thus, the PBS maps polarization-encoded qubits into distinct path modes, enabling conditional control between different degrees of freedom.

This highlights that the PBS conditionally directs the photon into a specific spatial output mode based on its polarization. Such a capability is essential in linear optical quantum computing, since it allows for the construction of interferometers, polarization-to-path encoding conversions, and multi-qubit entangling gates [10].

5 Entanglement and Interference

Quantum mechanics is often described as counterintuitive, but two of its most fascinating phenomena-*entanglement* and *interference*, form the foundation of photonic quantum computing. These effects are not only strange in comparison to classical physics, but are very powerful, allowing us to build quantum states and operations that have no classical counterpart. In this section, we introduce these ideas through intuitive narratives, mathematical representations, and practical methods for generating and using entangled photons.

5.1 Introduction to Entanglement

Imagine Alice and Bob, each holding a coin. In a classical world, someone could secretly prepare the coins so that they are always opposite. Whenever Alice flips her coin and sees heads, Bob is guaranteed to see tails. If she sees tails, Bob must see heads. This correlation feels remarkable, but is not mysterious- it arises because the coins were prepared with a hidden rule. The outcomes are predetermined, even if Alice and Bob do not know them until the coins are flipped.

Quantum entanglement goes beyond this type of hidden preparation, revealing a connection that has no classical counterpart. Suppose instead that the two coins are prepared in a quantum state that is a superposition of both possibilities:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |T\rangle_B + |T\rangle_A |H\rangle_B)$$

In this situation, before either coin is observed, neither Alice's nor Bob's coin has a definite face. They both exist in a blend of heads and tails. The moment Alice looks at her coin and finds heads, Bob's coin is *instantly* tails, even if he is far away. If Alice sees tails, Bob's coin collapses to heads. The key difference is that no hidden list of outcomes was assigned in advance- the definite results only emerge when a measurement is made.

A similar analogy works with dice. Suppose Alice and Bob each hold a die. Classically, one could rig the dice so that their numbers always add up to six. When Alice rolls and gets a "2," Bob is certain to find a "4." The correlation exists, but it is fully explained by hidden preparation. In the quantum version, the two dice are entangled in a state that is a superposition of all possibilities that sum to six:

$$|\Psi\rangle = \frac{1}{\sqrt{5}}\big(|1,5\rangle + |2,4\rangle + |3,3\rangle + |4,2\rangle + |5,1\rangle\big)$$

When Alice rolls her die, she might obtain a 3, and at that very moment Bob's die collapses to 3. If Alice sees a 5, Bob's die is instantly fixed at 1. The outcomes are not determined in advance, but instead arise from the structure of the shared quantum state itself.

This is the essence of entanglement: the results are perfectly correlated, but not in a way that any classical hidden mechanism can reproduce [37].

When we consider two photons, the combined system lives in a tensor product space. A general two-qubit state can be written as:

$$|\Psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,$$

where $|01\rangle$ means the first photon is in state $|0\rangle$ while the second photon is in state $|1\rangle$.

If this state can be factored into the form:

$$|\Psi\rangle = |\phi\rangle \otimes |\chi\rangle\,,$$

then the photons are independent. However, if no such factorization is possible, then the photons are entangled. For example,

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(|01\rangle + |10\rangle\big)$$

is an entangled state because it cannot be written as a product of two single-qubit states. Measuring one photon immediately gives us information about the other.

Hence, to "entangle photons" means to prepare them in a quantum state where their polarizations, paths, or other properties are correlated in a way that goes beyond classical statistics [38]. Unlike ordinary correlations, quantum entanglement creates correlations that cannot be explained by any hidden predetermined values. This universal correlation is the resource that makes quantum communication and quantum computing possible [2].

5.2 Generating Entangled Photons

While the concept of entanglement can be understood abstractly, it is of not much importance unless we generate and manipulate entangled particles in the laboratory for applications. In photonic quantum information science, photons are the most widely used carriers of entanglement because they are fast, robust against decoherence, and relatively easy to control with optical elements [16, 39]. But how can one actually create two photons that share such a property?

Instead of assigning each photon its own state, we are forced to describe the pair with a single, joint wavefunction. The correlations encoded in this shared state are what make them entangled [16].

There are various ways in which a single photon can be generated, each suitable for different applications. Some of the most popular methods are spontaneous parametric down-conversion (SPDC), spontaneous four-wave mixing (SFWM), trapped atom/ion sources, quantum dots and diamond-NV centres [39]. Each technique guarantees the conservation of energy and momentum, which in turn ensures strong correlations. Of these methods, SPDC is the most widely used method for generating entangled (∼99.5% fidelity) single photon pairs [40]. It has various applications in testing the fundamentals of quantum mechanics, quantum information processing, photonic quantum computing, etc. [30].

5.2.1 Spontaneous Parametric Down-Conversion (SPDC)

One of the most common techniques of producing entangled single-photons is *spontaneous parametric down-conversion* (SPDC). In this process, a high-energy photon from a laser (called the pump photon) enters a non-linear crystal, such as beta-barium borate (BBO). Inside the crystal, there is a small probability that the pump photon will "split" into two lower-energy photons, called the *signal* and *idler* photons. Energy and momentum conservation ensure that:

$$\omega_p = \omega_s + \omega_i$$
$$\vec{k}_p = \vec{k}_s + \vec{k}_i$$

where ω are the frequencies and $\vec{k}$ are wave vectors of the pump (p), signal (s), and idler (i) photons. These are known as *phase-matching conditions*, leading to constructive interference and efficient conversion [41].

In a carefully aligned crystal, the down-converted photons can emerge in polarization-entangled pairs, such as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|H\rangle_s |H\rangle_i + |V\rangle_s |V\rangle_i \big).$$

This is known as the Bell state $|\Phi^+\rangle$. Experiments with SPDC typically place two non-linear crystals at perpendicular orientations, ensuring that photons emerge in superpositions of horizontal and vertical polarizations, thus generating entanglement [30].

If we assume the wavelength of the signal and idler photons to be similar, we require the same

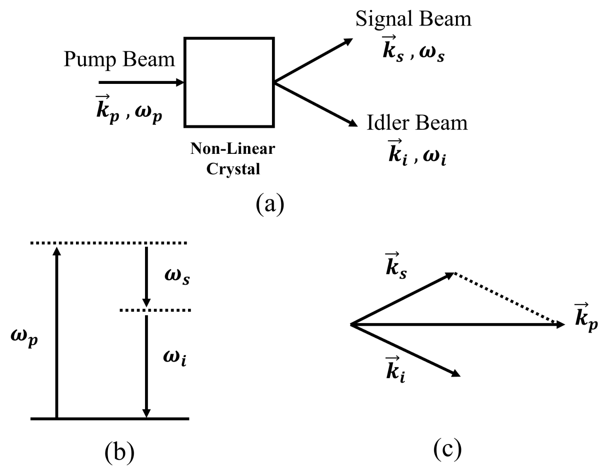


Figure 11: (a) The generation of signal-idler photon pair from the passing of a pump source through a non-linear crystal (b) Conservation of energy (c) Conservation of Momentum

refractive index for pump, signal, and idler photons. For this reason, birefringent (anisotropic, i.e., physical properties depend on direction) crystals are used. When light enters such a crystal, it splits into two rays with different polarization directions and different refractive indices as:

- **Ordinary (o) ray:** Polarization direction is perpendicular to the optical axis, having a constant refractive index n_o.
- **Extraordinary (e) ray:** Polarization direction is not perpendicular to the optical axis, and the refractive index varies with direction, denoted as $n_e(\theta)$.

Note that the *optical axis* is a special direction inside a birefringent crystal defined by the internal symmetry and structure of its atoms [42].

Type-I SPDC

In the Type-I SPDC process, the polarizations of signal and idler photons are the same, and perpendicular to that of the pump laser. For example, $e \rightarrow o + o$. A commonly used uniaxial birefringent crystal for this is β-Barium Borate (BBO).

Phase matching is achieved by rotating the crystal axes until the refractive indices satisfy the condition $n_e = n_o$. This allows efficient conversion from pump photons (typically at $\lambda_p \approx 405\,\text{nm}$) to signal-idler pairs (each typically at $\lambda_{s,i} \approx 810\,\text{nm}$).

Thicker crystals provide a higher pair-generation rate per unit pump power, since there is more volume for light to interact with the crystal. Depending on the exact phase-matching parameters, signal and idler photons can leave the crystal in either:

1. **Collinear geometry:** The pair leaves in the same direction as the pump beam. This results in high photon collection efficiency into a single optical fiber and greater spatial overlap, which is useful for integrated photonic circuits.

2. **Non-collinear geometry:** The photon pairs leave at different angles with respect to the pump beam. They form rings (2D projection) or cones (3D, especially in the case of Type-II SPDC) around the pump axis, and are spatially separated.

Type-II SPDC

In the Type-II SPDC process, the signal and idler photon pairs have orthogonal polarizations, and the pump can be either ordinary or extraordinary.
For example, $e \rightarrow o + e \qquad \text{or} \qquad o \rightarrow o + e$

This process requires careful tuning of the crystal orientation and temperature control to achieve phase matching. In the non-collinear case, the output photons emerge on two separate cones, one for each polarization. These cones intersect at two points, and it is exactly at these intersection points that polarization-entangled photons are generated [41].

5.3 Quantum Interference

Interference is one of the most striking and counterintuitive phenomena in physics. It occurs when two or more waves overlap in space and time, producing a resultant wave whose intensity depends on the relative phases of the individual waves. Classical examples of interference include the rippling patterns produced when water waves intersect or the bright and dark fringes in Thomas Young's double-slit experiment with light [34]. In each of these cases, the basic principle is that wave amplitudes add, and the observable intensity pattern is proportional to the square of this sum of amplitudes.

In optics, classical interference can be described as the superposition of electric-field amplitudes. Suppose that two coherent beams of light meet at a point. If their relative phase difference is zero, constructive interference occurs, and the intensity is maximized. If the relative phase is π, destructive interference takes place and the intensity vanishes. The constructive and destructive interference patterns arise from these simple rules of addition. Importantly, in classical interference, the underlying waves are deterministic and continuous. Even though one may not know their precise amplitudes at all points, the physics is fully governed by Maxwell's equations [43].

At the quantum level, interference arises in a subtler way, involving probability amplitudes rather than physical waves alone. Light is composed of photons that are quantized excitations of the electromagnetic field. Unlike classical waves, photons exhibit both particle-like and wave-like characteristics. When considering interference in quantum optics, it is not merely the classical electric fields that interfere, but rather the probability amplitudes associated with different quantum states of the photons. This distinction, though subtle, has remarkable consequences.

To get a feel of this, consider a single photon sent into a double-slit apparatus. Classically, one might expect the photon to pass through one slit or the other. However, experiments reveal that if both slits are open and no attempt is made to measure which slit the photon traverses, the photon interferes with itself, producing the same interference fringes as if it were a wave. The pattern only emerges after many photons are detected, but each individual photon carries the imprint of quantum interference [44, 45]. In contrast, if we attempt to detect which slit

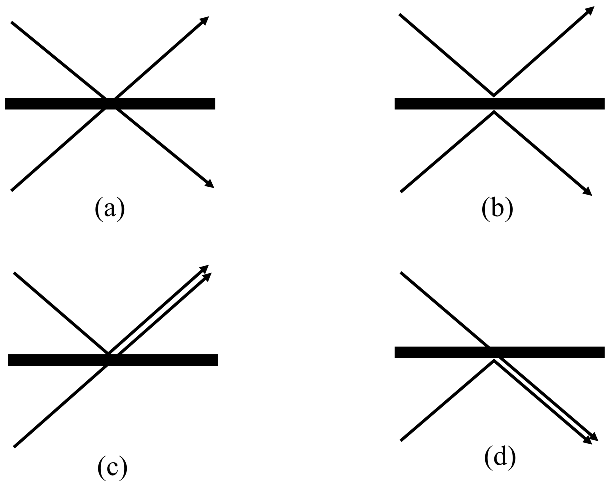


Figure 12: The HOM effect says that when two identical photons are incident at the 50:50 BS at the same time, cases (a) and (b) never occur, whereas there is a 50% probability that either of cases (c) or (d) occurs.

the photon passed through, the interference pattern vanishes, and we are left with a classical particle-like distribution. This is a hallmark of quantum interference: It arises not from physical waves that overlap in space but from the superposition of probability amplitudes associated with indistinguishable quantum paths.

5.3.1 The Hong-Ou-Mandel (HOM) Effect

The Hong-Ou-Mandel effect is one of the most striking demonstrations of quantum interference. In this experiment, two identical photons enter a 50:50 beam splitter from different input ports. Classically, we might expect each photon to randomly exit from either output port, giving a 50% chance that they leave separately, each taking a different output port, and sometimes they both end up in the same port, with relative probabilities [46].

Instead, quantum mechanics predicts, and experiments confirm, that when the photons are perfectly indistinguishable in all degrees of freedom (polarization, frequency, arrival time, spatial mode), the probability that they exit in different ports completely vanishes. The two photons always “bunch” together and exit from the same port. The resulting output state is [10]:

$$|1\rangle_a |1\rangle_b \xrightarrow{BS} \tfrac{1}{\sqrt{2}}\big(|2\rangle_a |0\rangle_b + |0\rangle_a |2\rangle_b\big).$$

This means that both photons are found in mode a or both in mode b, but never one in each. The effect arises because the probability amplitudes for the photons taking different outputs cancel each other perfectly. The experimental signature is a *dip in the coincidence count* when the photon arrival times are matched, known as the *HOM dip* [47, 48].

This two-photon interference is a purely quantum effect with no classical analog and is the basis for many entangling gates in linear optical quantum computing [14].

Photon Bunching and Indistinguishability

The HOM effect illustrates the concept of *bosonic bunching*—the tendency of identical bosons (such as photons) to occupy the same state. Indistinguishability is crucial: if the photons differ in polarization, frequency, or arrival time, the interference pattern diminishes. Achieving high visibility in HOM interference is therefore an experimental benchmark for the quality of photon sources in quantum optics [49].

5.4 EPR Paradox and the Bell States

Imagine two particles created in a single quantum process and then sent far apart- say, one to Alice on Earth and the other to Bob in a spaceship somewhere in outer space. Quantum mechanics tells us that their properties can remain mysteriously linked, no matter the distance. If Alice measures her particle and finds it in a particular state, Bob's particle instantaneously assumes a correlated state. This idea, as elegant as it is counterintuitive, has troubled some of the greatest minds in physics.

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen (EPR) formalized this intuition as the famous **EPR paradox** [50]. They argued that such instantaneous correlations, called "spooky action at a distance" by Albert Einstein, appeared to violate the principle of locality, which asserts that nothing can travel faster than light. If quantum mechanics were a complete theory, measuring one particle could somehow instantaneously affect the other. EPR were unsatisfied with this and proposed an alternative: perhaps the outcomes of measurements were already determined by '*hidden variables*' unknown to quantum theory. It meant that the correlations are real but predetermined, and quantum mechanics provides only a statistical description of reality.

This paradox raises a fundamental question: "Are the strange correlations predicted by quantum mechanics real properties of nature, or constructions of an incomplete theory?" For decades, this was largely a philosophical debate. The issue remained unresolved until John Bell, in the 1960s, formulated precise mathematical inequalities—known as **Bell's inequalities** [37], that could experimentally distinguish between local hidden-variable theories and the predictions of quantum mechanics.

Bell showed that if hidden variables for locality exist, then the results of measurements on two particles could only be correlated to a certain extent. In other words, there is a maximum amount of correlation that is allowed by any local hidden-variable theory. These limits are expressed as mathematical bounds, called Bell's inequalities. But quantum mechanics predicts correlations that can violate these bounds. From Aspect's 1980s experiments [38, 51, 52] to contemporary loophole-free tests [49, 53], experiments confirm that entanglement is a genuine, non-classical feature of nature.

At the heart of these experimental tests are the Bell states, special two-qubit states that represent the simplest and most symmetric forms of maximal entanglement. They form a complete orthonormal basis for the two-qubit Hilbert space, and each encodes correlations that cannot be separated into independent states of the two qubits. The four canonical Bell states are:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle),$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

These states form an orthonormal basis for two-qubit systems and are fundamental resources in quantum communication protocols such as quantum teleportation and superdense coding.

5.4.1 Generating a Bell State

One of the simplest and most illustrative examples of entanglement generation is the preparation of *bell states* using a quantum circuit composed of a Hadamard gate and a controlled-NOT (CNOT) gate [2]. Starting with two qubits initialized in the computational basis state $|00\rangle$, the Hadamard gate is first applied to the control qubit. This transforms the state into a superposition,

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle .$$

A subsequent CNOT gate, with the first qubit as control and the second as target, correlates the states of the two qubits such that the overall state becomes

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

This is one of the four maximally entangled Bell states. By choosing different initial states or by applying additional single-qubit gates, the other Bell states- $|\Phi^-\rangle$, $\left|\Psi^+\right\rangle$, and $|\Psi^-\rangle$ can be generated. This simple circuit demonstrates how entanglement can be systematically engineered from unentangled inputs, forming a fundamental resource for quantum teleportation, superdense coding, and a variety of quantum algorithms.

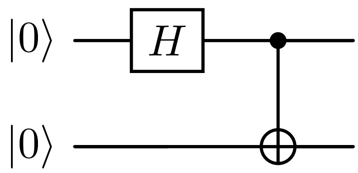


Figure 13: Quantum circuit for generating the Bell state $\left|\Phi^+\right\rangle$ from the initial state $|00\rangle$. A Hadamard gate on the first qubit followed by a CNOT entangles the two qubits.

Thus, entanglement and interference are not separate phenomena but deeply interconnected. Entanglement relies on interference at the level of probability amplitudes across multiple particles, while interference often manifests entanglement when photons become indistinguishable. Together, they provide the essential building blocks for photonic quantum technologies, enabling secure communication, quantum-enhanced metrology, and scalable quantum computation.

6 Optical Realization of Deutsch Algorithm

6.1 Introduction

This is one of the simplest examples of a quantum algorithm, yet it beautifully illustrates the fundamental advantages of quantum computation over classical methods [2]. To understand its significance, let us start with the classical problem it solves.

Consider a black-box boolean function $f : \{0, 1\} \to \{0, 1\}$. This function takes a single bit as input and outputs either 0 or 1. Without knowing the details of f, we have to find whether f is *constant* (the same output for both inputs 0 and 1) or *balanced* (different outputs for 0 and 1).

Function	$f(0)$	$f(1)$	**Type**
$f_1(x)$	0	0	Constant
$f_2(x)$	0	1	Balanced
$f_3(x)$	1	0	Balanced
$f_4(x)$	1	1	Constant

Table 3: All possible functions $f(x)$ for Deutsch's problem.

Classically, the solution requires two evaluations of the function in the worst case: we must check both $f(0)$ and $f(1)$ to be certain whether it is constant or balanced. If we only check one input, we cannot know whether the function is balanced or not.
For example, suppose we call the function with input 0 and obtain the output $f(0) = 1$. From the table of possible functions (Table 3), this tells us we are either in the case of $f_2(x)$ or $f_4(x)$. However, we would still need to call the function a second time, with input 1, in order to distinguish between these two cases.

This is a simple problem, but it highlights the limitation of classical computation: each input must be evaluated individually.

6.2 Quantum Parallelism

Quantum mechanics offers a powerful solution to this through the principle of *quantum parallelism* [2]. In the quantum setting, we can prepare a superposition of input states, effectively allowing the function to be evaluated for both inputs simultaneously. For the Deutsch problem, we represent the input qubit in the superposition state-

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

By passing this qubit along with an auxiliary qubit- through a quantum oracle $\boldsymbol{U_f}$ that implements the function f, the system encodes information about both inputs at once. This is the essence of quantum parallelism: a single operation processes multiple inputs simultaneously in a coherent superposition.

6.3 The Deutsch Algorithm

Note that we rely on the following key principles [2]:

- **Superposition:** We prepare a qubit in a superposition state and then apply $f(x)$ on it. This allows us to evaluate $f(x)$ for both inputs at the same time.
- **Phase Kickback:** It essentially means that the phase $(-1)^{f(x)}$ is associated with the second qubit, but we have the freedom to associate it with the first qubit. So, the oracle operates on the second qubit only when $|x\rangle = |1\rangle$, and acts as an identity operator for $|x\rangle = |0\rangle$. This clever trick allows the information about $f(x)$ to be revealed with a single measurement.

Now let us understand this algorithm [54] step by step.

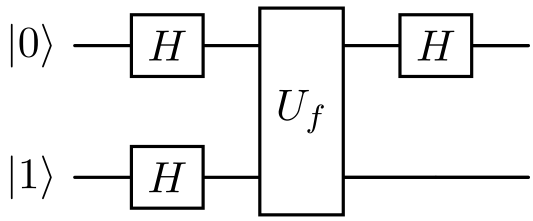


Figure 14: Quantum circuit for the Deutsch algorithm. The first qubit is initialized in $|0\rangle$, the second in $|1\rangle$. After applying Hadamard gates, the oracle U_f encodes the function, followed by a final Hadamard on the first qubit before measurement.

(1) Initialization. Start with two qubits: the data qubit initialized to $|0\rangle$ and the ancilla qubit to $|1\rangle$:

$$|\psi_0\rangle = |0\rangle \otimes |1\rangle = |0\rangle\,|1\rangle\,. \tag{3}$$

(2) Apply Hadamards (Superposition). Apply a Hadamard gate H to both qubits. The Hadamard acts as-

$$H\,|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H\,|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}, \tag{4}$$

so the state becomes

$$|\psi_1\rangle = (H\,|0\rangle) \otimes (H\,|1\rangle) \tag{5}$$

$$= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \tag{6}$$

$$= \frac{1}{2}\Big(\,|00\rangle - |01\rangle + |10\rangle - |11\rangle\,\Big) \tag{7}$$

$$= |+\rangle \otimes |-\rangle\,, \tag{8}$$

where $|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$.

(3) Oracle action and phase kickback. The oracle U_f encodes f via-

$$U_f\,|x,y\rangle = |x,\, y \oplus f(x)\rangle\,, \qquad x,y \in \{0,1\}. \tag{9}$$

On the ancilla $|-\rangle$, this induces *phase kickback*:

$$\begin{aligned} U_f\big(\,|x\rangle \otimes |-\rangle\,\big) &= \tfrac{1}{\sqrt{2}}\,U_f\big(\,|x,0\rangle - |x,1\rangle\,\big) &(10)\\ &= \tfrac{1}{\sqrt{2}}\big(\,|x,f(x)\rangle - |x,1\oplus f(x)\rangle\,\big) &(11)\\ &= (-1)^{f(x)}\;|x\rangle \otimes |-\rangle\,. &(12) \end{aligned}$$

Applying U_f to $|\psi_1\rangle$ gives

$$|\psi_2\rangle = U_f\,|\psi_1\rangle = \tfrac{1}{\sqrt{2}}\Big((-1)^{f(0)}\,|0\rangle + (-1)^{f(1)}\,|1\rangle\Big) \otimes |-\rangle\,. \tag{13}$$

(4) Final Hadamard on the data qubit. Apply H to the first qubit only. Using

$$H\left(\tfrac{|0\rangle+|1\rangle}{\sqrt{2}}\right) = |0\rangle\,, \qquad H\left(\tfrac{|0\rangle-|1\rangle}{\sqrt{2}}\right) = |1\rangle\,, \tag{14}$$

we distinguish two cases:

- If f is **constant**, then $f(0) = f(1)$ and the first-qubit state before H is $\pm\frac{|0\rangle+|1\rangle}{\sqrt{2}}$, which maps to $|0\rangle$.
- If f is **balanced**, then $f(0) \neq f(1)$ and the first-qubit state before H is $\pm\frac{|0\rangle-|1\rangle}{\sqrt{2}}$, which maps to $|1\rangle$.

Therefore, up to a global phase,

$$|\psi_{\text{out}}\rangle = |\,f(0) \oplus f(1)\,\rangle \otimes |-\rangle\,, \tag{15}$$

so a single measurement of the first qubit yields 0 for *constant* and 1 for *balanced*—with one oracle call.

Note: Superposition enables evaluating f on both inputs in parallel, while *phase kickback* transfers the information about f into a relative phase that the final Hadamard converts into a deterministic, classical bit. The controlled operation can be implemented with gates such as CNOT as part of the oracle construction.

6.4 Linear Optical Implementation of Deutsch's Algorithm

While the Deutsch algorithm is often presented abstractly in terms of qubits and unitary gates, its true power is revealed when we attempt to implement it on a physical platform. We have discussed how linear optical quantum computing (LOQC) stands out among the various architectures for quantum computation [10, 14]. In LOQC, qubits are encoded in different degrees of freedom of single photons—most commonly their polarization states $|H\rangle$ and $|V\rangle$ or spatial modes (path encoding). Quantum gates are realized using simple optical components such as beam splitters, mirrors, phase shifters, and polarizing elements [10, 47].

In the optical framework, the abstract idea of superposition becomes a physical reality: a single photon can be in more than one path or polarization mode at the same time, allowing it to represent different inputs to a function simultaneously. When these paths are brought together in a beam splitter or wave plate, the probability amplitudes of the photon interfere in a way that directly reflects the mathematical interference step in the algorithm. The outcome of this interference depends on whether the encoded function is constant or balanced, and a single measurement at the output is enough to reveal the answer—just as prescribed by the quantum circuit model, but now realized in a physical experiment [21, 48].

6.4.1 Initialization and Superposition

The experiment begins with the preparation of a single-photon source, which can be realized through spontaneous parametric down-conversion (SPDC) [42, 46]. One photon of the pair is detected to "herald" the presence of the other, ensuring that the experiment begins with a well-defined single photon source. This heralded photon is then sent into the optical circuit. To represent the two qubits required by the Deutsch algorithm, different physical properties of the photon are used: its polarization and its spatial mode (path). The polarization qubit is initialized in the horizontal state $|H\rangle \equiv |0\rangle$, while the path degree of freedom defines the second qubit. Choosing one of the available paths corresponds to the logical state $|1\rangle$. Together, the initial two-qubit state of the system is defined as $|01\rangle$.

The Hadamard operation on the polarization qubit is implemented using a half-wave plate (HWP) at 22.5°. Acting on $|H\rangle \equiv |0\rangle$, this creates a superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

For the path qubit, a 50:50 beam splitter (BS) performs the Hadamard-like operation [47]. So the state of the system becomes

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|H\rangle - |V\rangle}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}[|00\rangle - |01\rangle + |10\rangle - |11\rangle].$$

This creates a coherent superposition of the two possible paths, allowing the photon to effectively explore both inputs to the oracle at the same time—a direct physical realization of quantum parallelism [21].

6.4.2 Oracle Implementation

In this optical realization of the Deutsch algorithm, the oracle operation is implemented by inserting half-wave plates (HWPs) into specific arms of a Mach–Zehnder interferometer [48]. The required oracle transformation maps the two-qubit state as

$$|x, y\rangle \;\mapsto\; |x,\, y \oplus f(x)\rangle$$

where x is the path (0 for the lower arm, 1 for the upper arm), y is the polarization, and $f(x)$ represents the Boolean function to be tested.

A HWP oriented at 45° acts as a Pauli-X gate on polarization, flipping horizontal $|H\rangle$ and vertical $|H\rangle$ states [33]. When such a plate is placed in a given interferometer arm, it applies this flip only if the photon travels along that path. Thus, the choice of HWP placement directly encodes the Boolean function. For a constant function, either no HWPs (constant 0) or HWPs in both arms (constant 1) are used, ensuring that polarization is treated identically regardless of path. For a balanced function, a HWP is introduced in only one arm, such that the polarization is flipped if and only if the photon travels through that path. This yields the conditional transformation characteristic of balanced functions.

In this way, the oracle is realized entirely through simple polarization control elements. The placement of HWPs determines whether the implemented function is constant or balanced, while the subsequent interference at the second beam splitter translates this information into measurable detector outcomes [48, 55]. This approach highlights the elegance of optical quantum information processing, where logical operations can be encoded through straightforward physical manipulations of light.

6.4.3 Final Hadamard and Measurement

After the oracle, another Hadamard gate is applied to the polarization qubit using a HWP, while the second beam splitter performs interference on the path qubit [47].

The final measurement is made in the polarization-basis. If the function is constant, the photon will always be detected in the horizontal polarization channel $|H\rangle$, while if the function is balanced, the photon will be detected in the vertical polarization channel $|V\rangle$. Mathematically, the measurement outcome of the polarization qubit encodes the result of the Deutsch problem:

$$\text{Output} = \begin{cases} |0\rangle & \text{if function is constant,} \\ |1\rangle & \text{if function is balanced.} \end{cases}$$

6.4.4 Experimental Considerations

It is important to recognize that in the scheme described above, the oracle must be physically modified depending on whether the function is constant or balanced. For constant functions, the oracle action is straightforward: either the photon's path is left unchanged, or both paths are swapped in a way that is independent of polarization. For balanced functions, however, the oracle must be polarization-dependent- when the photon is in the state $|H\rangle$, its path is left untouched, while in the state $|H\rangle$ the path qubit is flipped, meaning that a photon in the upper arm is redirected to the lower arm, and vice versa. In practice, this is achieved by placing half-wave plates or phase plates in the optical paths to introduce the required transformations.

Although this approach doesn't seem to capture the power of Deutsch's algorithm since the experimenter needs to reconfigure the setup for different oracle choices, it is introduced here for pedagogical reasons. It shows in the clearest possible way how a simple optical element can be used to "program" the oracle and how polarization can act as a control on the photon's path.

More advanced schemes, however, overcome this limitation by using entanglement as a resource. Instead of physically changing the optical elements for each oracle, one can prepare entangled photon pairs in which polarization and path degrees of freedom are correlated. In such a setup, the distinction between constant and balanced functions is revealed by the correlations observed in coincidence measurements, without the need to reconfigure the hardware. The approach described in the attached reference illustrates this more sophisticated method. In this way, the simple optical implementation provides an intuitive starting point, while entanglement-based schemes demonstrate how quantum resources enable more elegant and powerful realizations of algorithms like Deutsch's [56].

In practice, high interference visibility and coherence are crucial. The optical path lengths must be stabilized to within the photon coherence length, and polarization drift must be minimized. Integrated photonic circuits, such as silica-on-silicon waveguides, provide an excellent solution by offering stable interferometers and precise phase control. Such integrated platforms have been successfully used to implement the Deutsch algorithm and other quantum logic circuits. [55, 57, 58]

Thus, the linear optical implementation of Deutsch's algorithm illustrates how fundamental quantum computational speed-up can be physically realized. The mapping of abstract quantum gates onto polarization and path transformations demonstrates the practicality of photonic quantum computing using only linear optical elements.

7 Conclusion

In this review, we have outlined the essential foundations of photonic quantum computing with a focus on pedagogy and accessibility. Starting from the representation of qubits in photons through polarization, dual-rail, and time-bin encodings, we described how optical elements such as beam splitters, phase shifters, and polarizing beam splitters implement single-qubit logic. We then highlighted the importance of quantum interference and entanglement, illustrated through spontaneous parametric down-conversion and the Hong-Ou-Mandel effect, as the key physical resources enabling non-classical correlations and computational power. These discussions provide newcomers with the conceptual background necessary to appreciate how light can be harnessed for both quantum communication and computation.

Recent research has significantly advanced the field, moving from bulk-optics demonstrations to scalable, integrated photonic platforms. Photonic chips fabricated in silicon, silicon nitride, and other materials allow for the miniaturization of complex optical circuits, improving stability and scalability compared to free-space implementations [59]. Experiments in boson sampling [60] have provided strong evidence of photonic systems achieving quantum advantage for specific sampling problems [61, 62], marking an important milestone toward near-term photonic quantum devices. Meanwhile, hybrid architectures that couple photons with matter-based systems, such as quantum dots, trapped ions, or atomic ensembles, are being pursued to realize deterministic entanglement, quantum repeaters, and long-distance quantum networks [63]. These developments suggest a future where photonic quantum computing and quantum communication converge in the realization of a global quantum internet [64, 65].

Nevertheless, challenges remain. Scalable implementations require deterministic two-qubit gates, efficient error correction, and high-quality single-photon sources and detectors. Current approaches, including multiplexed photon generation, improved nonlinear materials, and error-resilient optical architectures, aim to overcome these limitations [13, 66]. As the field progresses, photons are expected to remain a central resource—not only as robust carriers of quantum information but also as building blocks for scalable, fault-tolerant quantum technologies.

By presenting the underlying principles in a clear and accessible way, this pedagogical review serves as a gateway for students and early researchers to enter the field of photonic quantum computing. With both foundational understanding and awareness of current trends, readers are equipped to engage with ongoing research and contribute to the rapid developments shaping the future of quantum technologies.

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